Adaptive Policy Optimization for Municipal Solid Waste Segregation using Coupled Agent-Based Modeling and Deep Reinforcement Learning

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***Abstract*—Effective municipal solid waste (MSW) segregation is a critical sustainability bottleneck. Applying Deep Reinforcement Learning (DRL) to optimize policy in multi-agent human environments faces the sparse reward problem due to cultural inertia. We introduce a socio-technical framework coupling an Agent-Based Model (ABM) with a Heuristic-Guided DRL (HuDRL) agent for predictive governance. Using Bacolod, Lanao del Norte as a testbed, we optimize resource allocation via a Theory of Planned Behavior (TPB) cognitive model. Global Sensitivity Analysis proves that internal household friction accounts for 79% of system variance. To navigate this non-convex space, the HuDRL agent utilizes action masking and a targeted amplification algorithm. Simulations reveal that while standard policies stagnate, the HuDRL agent converges on a "Sequential Saturation" strategy—concentrating funding into single nodes until tipping points are breached. This approach proves vastly superior to equitable distribution, directly advancing Sustainable Development Goals 11, 12, and 16.**

***Index Terms*—Agent-Based Modeling, Deep Reinforcement Learning, MSW Management, Sparse Reward Problem, Theory of Planned Behavior.**

I. Introduction

Sustainable society requires systems capable of optimizing complex human environments. Socio-economic systems are characterized by bounded rationality and cultural inertia, creating non-convex policy spaces where equitable resource distribution—a common human bias—is often suboptimal. From a computational view, standard DRL faces a severe "Sparse Reward Problem" [2] because human behavior shifts require sustained pressure before state changes occur.

This study uses Municipal Solid Waste (MSW) management as a testbed. In the Philippines, despite the mandates of R.A. 9003, Local Government Units (LGUs) struggle with the "MRF Paradox": while recovery facilities increased by 8.7% in 2024, illegal dumpsites concurrently surged by 84% [3]. We propose a coupled ABM and DRL architecture integrating TPB to simulate structural friction. A HuDRL agent representing municipal authority then optimizes budget allocations to overcome behavioral inertia within strict fiscal realities.

II. Related Work

A. Behavioral Friction and Tipping Points

TPB posits that intention is a function of Attitude, Subjective Norms, and Perceived Behavioral Control [4]. However, environmental compliance is frequently obstructed by the "Cost of Effort" [5]. Sustained change relies on tipping points; once a minority reaches $\sim$25%, it can trigger cascades across the majority [6], creating a "Social Norm Shield" where behavior resists decay at compliance $\geq$70% [7].

B. Algorithmic Policy Design

Traditional optimization fails to capture non-linear social hysteresis [8]. While research exists for MSW logistics [9], behavioral bottlenecks remain understudied. ABMs offer virtual laboratories, but manual testing is labor-intensive. By formalizing environments as a Markov Decision Process (MDP), neural networks can discover adaptive strategies, provided they utilize heuristics and action masking to respect fiscal boundaries [2].

III. System Architecture

The simulation is bounded to Bacolod, Lanao del Norte, containing 4,022 Household Agents across seven barangays. The architecture uses a dual-clock temporal resolution: households act daily (micro-ticks), while the Mayor Agent optimizes budgets quarterly (90-day epochs) over a 3-year horizon. Meso-level Barangay Agents aggregate micro-data into a 17-dimensional state vector, reducing computational complexity to $\mathcal{O}(N)$.

IV. The Multi-Agent Environment

A. Household Agents and Cognitive Weights

Household utility (U_{seg}) is governed by a weighted TPB equation:

$$U_{seg} = (w_A(t)A)+(w_{SN}(t)SN_{loc})+(w_{PBC}(t)PBC)-C_{Net}+\epsilon \quad (1)$$

where C_{Net} represents internal household friction (sorting and storing) weighed against incentives and fines [10]. If local compliance breaches 70%, behavior internalizes, hard-locking $w_{sn}(t)$ at 0.90.

B. Enforcement and Local Administrative Nodes

Enforcement agents execute stochastic random walks with a cost of PHP 12,000 per macro-enforcer. Barangay Agents execute a Cascading Incentive Protocol, drawing from local allotments and municipal augmentation. Exhausted budgets trigger a "Perceived Unfairness" penalty, degrading household motivation [5].

V. The Deep Reinforcement Learning Agent

A. MDP Formulation and Training

The state space is a 17-dimensional vector representing compliance, budget exhaustion, and political capital. The 21-dimensional action space controls three fiscal instruments per barangay. We use PPO with a Temperature-Scaled Softmax ($\tau = 5.0$) to ensure the total budget (PHP 375,000/quarter) is fully allocated. Training details are in Table I.

TABLE I
Training Configuration

Parameter	Value	Parameter	Value
Learning Rate	0.0003	Horizon	1024
Batch Size	128	γ	0.99
Architecture	MlpPolicy	Seeds	5

B. Reward Shaping

The composite reward R_t uses a +200 point $R_{Jackpot}$ for breaching the 70% threshold and a P_{Waste} quadratic penalty for redundant funding in "graduated" barangays. A logarithmic Φ_{cap} penalty prevents political collapse if approval drops below 25%.

VI. Results and Discussion

A. Sensitivity Analysis

Sobol GSA (Table II) reveals that C_{Effort} is the dominant driver of compliance ($S_1 = 0.79$), while Attitude Weight exhibits near-zero sensitivity. This proves that structural support is more critical than information campaigns for waste segregation.

TABLE II
Sobol Sensitivity Indices

Input Parameter	First-Order (S_1)	Total-Order (S_T)
Attitude Weight (w_A)	0.00	0.10
Cost of Effort (C_{Effort})	**0.79**	**1.06**

B. Comparative Analysis

HuDRL outperformed Vanilla PPO (Reward: 10551.7 vs 1728.6) by discovering a "Sequential Saturation" strategy. As shown in Fig 1, the AI concentrated ≥60% of funds onto single barangays to force social tipping points. Once compliance reached 70%, the "Social Norm Shield" maintained the state, allowing the agent to pivot funds to the next bottleneck. This method reached 90.45% compliance, whereas equitable distribution stagnated at 30.18%.

C. Limitations

Validity is bounded by the calibration to a single 4th-class municipality. Baseline psychological parameters were literature-seeded, and household agents lack deep demographic heterogeneity beyond income-sensitivity. Future work should explore MARL for commercial actors and 10-year horizons to test habit permanence.

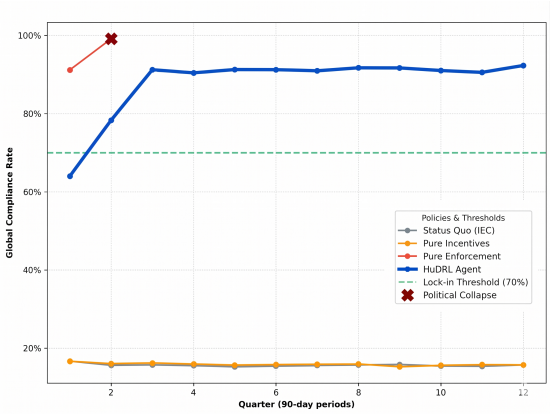

Fig. 1: Compliance Rate Performance

VII. Conclusion

HuDRL proves that building resilient societies requires abandoning equitable resource dilution in favor of targeted saturation. By formalizing policy as an MDP, LGUs can replace trial-and-error budgeting with precision-targeted mathematical strategies.

Acknowledgment

Source code available at `github.com/KINGSTING/THESIS_ABMxDRL`. Language refinement aided by Gemini Pro.

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